Definition : A right circular cone is a solid generated by the revolution of a right angled triangle about one of the sides containing the right angle as axis.

Formation of a Right Circular Cone - Demonstration

In Fig. 7.10, if the right triangle ABC revolves about AB as axis, the hypotenuse AC generates the curved surface of the cone represented in the diagram. The height of the cone is the length of the axis AB, and the slant height is the length of the hypotenuse AC.

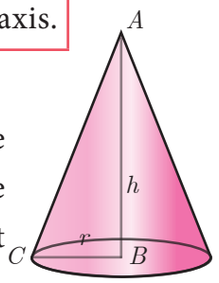


Fig. 7.10

Surface area of a right circular cone

Suppose the surface area of the cone is cut along the hypotenuse AC and then unrolled on a plane, the surface area will take the form of a sector ACD, of which the radius AC and the arc CD are respectively the slant height and the circumference of the base of the cone.

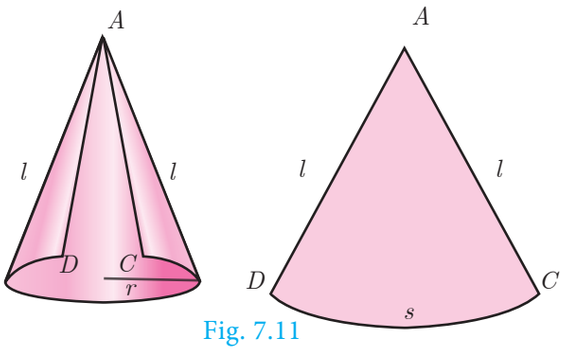


Fig. 7.11

Here the sector of radius 'l' and arc length 's' will be similar to a circle of radius l.

(i) Curved surface area

$$\frac{\text{Area of the sector}}{\text{Area of the circle}} = \frac{\text{Arc length of the sector}}{\text{Circumference of the circle}}$$

$$\text{Area of the sector} = \frac{\text{Arc length of the sector}}{\text{Circumference of the circle}} \times \text{Area of the circle}$$

$$= \frac{s}{2\pi l} \times \pi l^2 = \frac{s}{2} \times l = \frac{2\pi r}{2} \times l \qquad (\because s = 2\pi r)$$

$\therefore$ Curved Surface Area of the cone = Area of the Sector = πrl sq. units.

C.S.A. of a right circular cone $= \pi rl$ sq. units.

Thinking Corner

1. Give practical example of solid cone.
2. Find surface area of a cone in terms of its radius when height is equal to radius.
3. Compare the above surface area with the area of the base of the cone.

Activity 1

(i) Take a semi-circular paper with radius 7 cm and make it a cone. Find the C.S.A. of the cone.

(ii) Take a quarter circular paper with radius 3.5 cm and make it a cone. Find the C.S.A. of the cone.

10th_Maths_Chapter 7_English.indd 274 12/3/2024 2:59:30 PM